Home Assignment 2

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Problem 1

(a)

We want to compute the marginal posteriors $f(\theta|\boldsymbol{\lambda}, \mathbf{t}, \boldsymbol{\tau})$, $f(\boldsymbol{\lambda}|\theta, \mathbf{t}, \boldsymbol{\tau})$ and $f(\mathbf{t}|\theta, \boldsymbol{\lambda}, \boldsymbol{\tau})$ and identify the distributions.

We use Bayes' theorem to see that

$$f(\theta|\boldsymbol{\lambda}, \mathbf{t}, \boldsymbol{\tau}) \propto f(\boldsymbol{\tau}, \boldsymbol{\lambda}, \mathbf{t}, \theta) = f(\boldsymbol{\tau}|\boldsymbol{\lambda}, \mathbf{t}) f(\boldsymbol{\lambda}|\theta) f(\mathbf{t}) f(\theta) \propto f(\boldsymbol{\lambda}|\theta) f(\theta)$$

where the prior on each intensity has the probability density

$$f(\lambda_i|\theta) = \theta^2 \lambda_i e^{-\theta\lambda_i}, \quad \lambda_i > 0$$

and where we have a hyperprior on θ given by the probability density

$$f(\theta) = \vartheta^2 \theta e^{-\vartheta\theta}, \quad \theta > 0,$$

with the hyperparameter ϑ fixed. Since the intensities λ_i are conditionally independent we get

$$f(\boldsymbol{\lambda}|\theta) = \prod_{i=1}^{d} \theta^2 \lambda_i e^{-\theta\lambda_i}$$

meaning that

$$f(\theta|\boldsymbol{\lambda}, \mathbf{t}, \boldsymbol{\tau}) \propto \prod_{i=1}^{d} \theta^2 \lambda_i e^{-\theta\lambda_i} \cdot \vartheta^2 \theta e^{-\vartheta\theta} \propto \theta^{2d+1} \exp\left(-\theta\left(\vartheta + \sum_{i=1}^{d} \lambda_i\right)\right)$$

which can be identified to be $\Gamma\left(2d+2, \vartheta + \sum_{i=1}^{d} \lambda_i\right)$ distributed.

In the same way of procedure we see that

$$f(\lambda_i|\theta, \mathbf{t}, \boldsymbol{\tau}) \propto f(\boldsymbol{\tau}|\boldsymbol{\lambda}, \mathbf{t}) f(\boldsymbol{\lambda}|\theta)$$

where

$$f(\boldsymbol{\tau}|\boldsymbol{\lambda}, \mathbf{t}) \propto \exp\left[-\sum_{j=1}^{d} \lambda_j (t_{j+1} - t_j)\right] \prod_{j=1}^{d} \lambda_j^{n_j(\boldsymbol{\tau})}$$

such that

$$f(\lambda_i|\theta,\mathbf{t},\boldsymbol{\tau}) \propto \left\{e^{-\lambda_i(t_{i+1}-t_i)}\lambda_i^{n_i(\boldsymbol{\tau})}\right\} \underbrace{\left\{\exp\left[-\sum_{j\neq i}\lambda_j(t_{j+1}-t_j)\right]\prod_{j\neq i}\lambda_j^{n_j(\boldsymbol{\tau})}\right\}}_{\text{constant w.r.t } \lambda_i} \cdot \left\{\theta^2\lambda_i e^{-\theta\lambda_i}\right\} \underbrace{\left\{\prod_{j\neq i}\theta^2\lambda_j e^{-\theta\lambda_j}\right\}}_{\text{constant w.r.t } \lambda_i}$$

$$\propto \lambda_i^{n_i(\boldsymbol{\tau})+1} e^{-\lambda_i(\theta+(t_{i+1}-t_i))}$$

which is identified to be $\Gamma\left(n_i(\boldsymbol{\tau})+2, \theta+(t_{i+1}-t_i)\right)$ distributed.

Finally, we see that

$$f(\mathbf{t}|\theta,\boldsymbol{\lambda},\boldsymbol{\tau}) \propto f(\boldsymbol{\tau}|\boldsymbol{\lambda},\boldsymbol{t})f(\mathbf{t}) \propto \exp\left(-\sum_{i=1}^{d}\lambda_i(t_{i+1}-t_i)\right)\prod_{i=1}^{d}\lambda_i^{n_i(\boldsymbol{\tau})}\cdot\prod_{i=1}^{d}(t_{i+1}-t_i)$$

which cannot be recognized to be a specific Gamma distribution meaning we cannot directly sample from this distribution, indicating the need for the MH-step.

(b)

To update $\mathbf{t}$, the *random walk proposal* was used. The algorithm was initialized with $\boldsymbol{\lambda}=\mathbf{1}$, $\theta=1$, and the breakpoints equally spaced in $[1851, 1963]$. The hyperparameter ϑ and tuning parameter ρ were varied in parts (d) and (e) respectively, with $\vartheta=1$ and $\rho=1$ chosen as a starting point to be improved upon.

The hybrid MCMC algorithm proceeds as follows at each iteration:

1. Sample θ from its full conditional using Gibbs:

$$\theta \sim \Gamma\left(2d+2,\ \vartheta+\sum_{i=1}^{d}\lambda_i\right) \tag{1}$$

2. For each interval $i=1,\ldots,d$, count the disasters n_i in $[t_i, t_{i+1})$ and sample λ_i using Gibbs:

$$\lambda_i \sim \Gamma\left(n_i+2,\ \theta+t_{i+1}-t_i\right) \tag{2}$$

3. For each interior breakpoint $i=2,\ldots,d$, propose a candidate using the random walk proposal:

$$t_i^* = t_i+\varepsilon, \quad \varepsilon\sim\text{Unif}(-R,R), \quad R=\rho(t_{i+1}-t_{i-1}) \tag{3}$$

Accept t_i^* with probability:

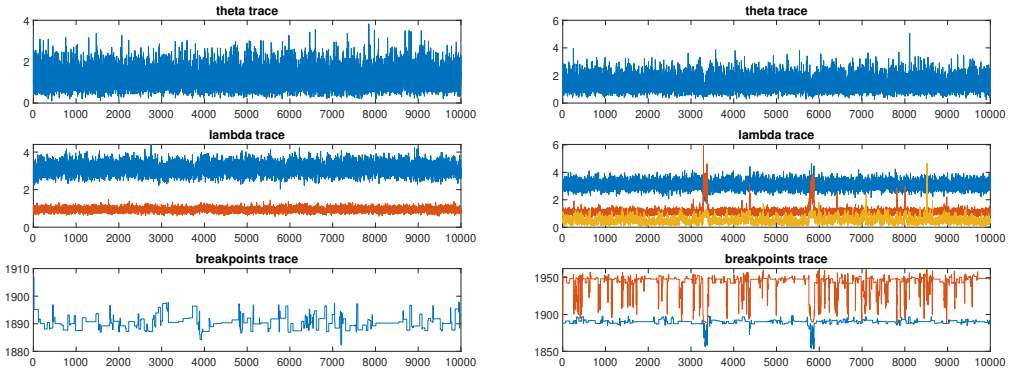
$$\alpha = \min\left(1,\ \frac{f(t^*|\theta,\boldsymbol{\lambda},\tau)}{f(t|\theta,\boldsymbol{\lambda},\tau)}\right) \tag{4}$$

4. Store θ, $\boldsymbol{\lambda}$, and $\mathbf{t}$.

Steps 1–4 are repeated for $N=10000$ iterations.

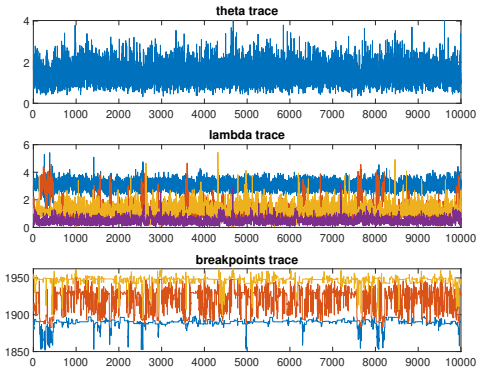
(c)

After running the hybrid MCMC algorithm for $1, 2, 3$ and 4 breakpoints, the parameter values θ_i and the intensities $\boldsymbol{\lambda}_i$ as well as the breakpoints $\mathbf{t}_i$, $i=1,\ldots,N$, were plotted and can be seen in Figure 1.

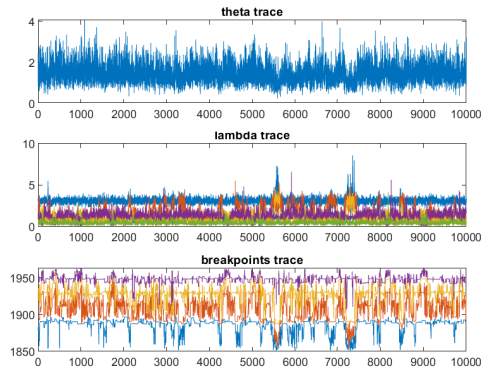


(a) One breakpoint

(b) Two breakpoints



(c) Three breakpoints



(d) Four breakpoints

Figure 1: Trace plots for θ, λ, and breakpoints for 1-4 breakpoints.

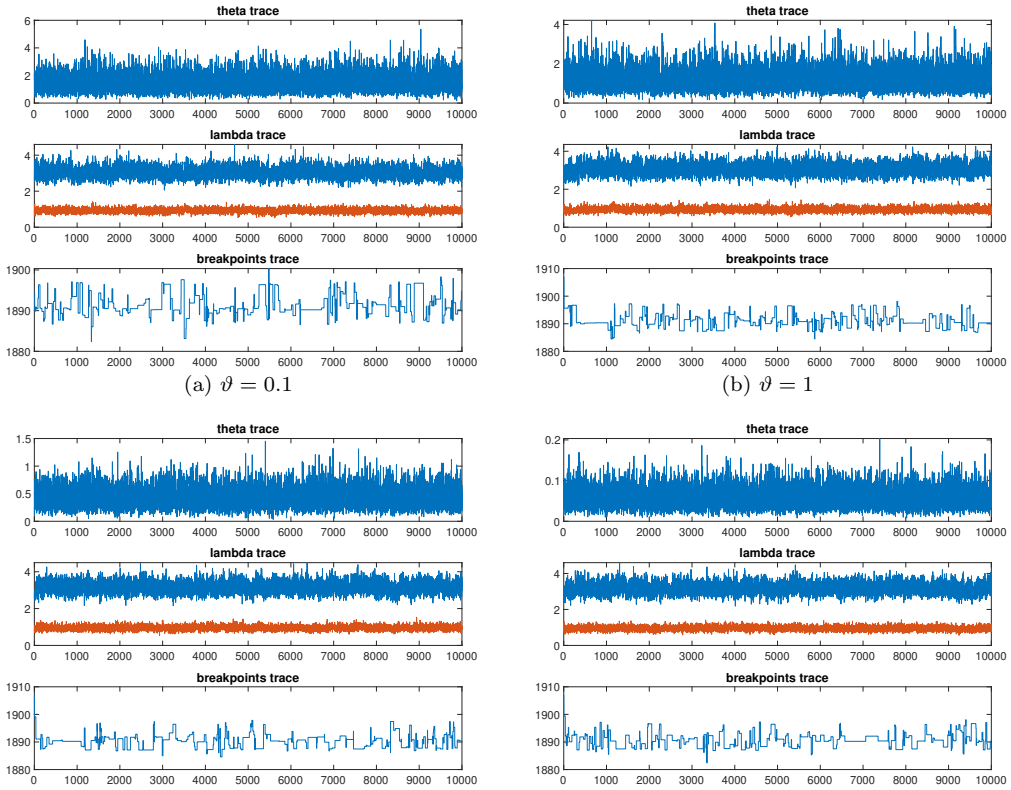
As we can see in the figures, as we increase the number of breakpoints, the more mixing we get between the breakpoints and between the intensities in the different intervals. This means that there is posterior uncertainty resulting in label switching between the intensities in the adjacent intervals, indicating that there might not be a need for a breakpoint to separate the intervals.

The only breakpoint without any mixing is then the special case with one breakpoint where there is no overlap between the interval intensities, as can be seen in Figure 1a. With one breakpoint around the 1890s, we get an average disaster intensity of 3.09 accidents/year before the 1890s and 0.93 accidents/year after. This suggests that around year 1890 there was an improvement made in safety regulation, effectively lowering the disaster intensity.

(d)

From part (c) we observe that the chain with one breakpoint mixes well and converges to a stable posterior, with the breakpoint consistently settling around 1890. For two or more breakpoints the chains exhibit poor mixing, with breakpoints jumping erratically across the full range $[1851, 1963]$, suggesting the data does not support more than one change point. Therefore all subsequent analysis is performed with one breakpoint.

We now want to see how the hyperparameter ϑ affects the posteriors. The traces for varying ϑ can be seen in Figure 2.



(a) $\vartheta = 0.1$ (b) $\vartheta = 1$

(c) $\vartheta = 10$ (d) $\vartheta = 100$

Figure 2: Trace plots for θ, λ, and 1 breakpoint

Firstly, we note that Figure 2b uses the same ingoing parameters as Figure 1a. In the figures, we can see that as we increase ϑ, the draws from the marginal posterior for θ decreases while the draws from the marginal posterior for $\boldsymbol{\lambda}$ stays roughly the same, giving also roughly the same breakpoint candidates. This falls in line with what was defined for the marginal posteriors, where for $\boldsymbol{\lambda}$, the draws are mostly driven by the data likelihood with slight regularization by θ through ϑ, whereas for θ the draws are mostly driven by the hyperprior depending on ϑ.

We can draw the conclusion that while the marginal posterior for θ is sensitive to ϑ, the marginal posterior for $\boldsymbol{\lambda}$ and $\mathbf{t}$ stays robust.

(e)

According to optimal scaling theory for random walk Metropolis-Hastings (Lecture 12 notes), the ideal acceptance rate is approximately 23%–44%. Our results show that ρ in the range 0.04–0.10 yields acceptance rates in this range, while the breakpoint chain moves freely and settles around 1890. For very small values such as $\rho \leq 0.01$, almost all proposals are accepted but the breakpoint barely moves each iteration, resulting in a slowly varying chain. For large values such as $\rho \geq 1$, nearly all proposals are rejected and the chain gets stuck. We therefore select $\rho = 0.05$ as a reasonable default. This gives the following:

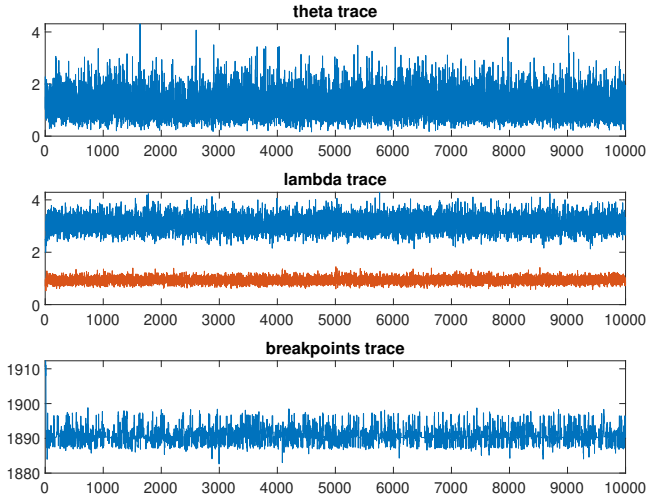


Figure 3: Trace plots for $\rho = 0.05$

This results in an acceptance rate of 0.3992 or $\approx 40\%$ which falls into the generally ideal range. We can also note that the marginal posteriors seem to be robust to ρ since the parameter $\boldsymbol{\theta}$ and the intensity $\boldsymbol{\lambda}$ does not change a lot when varying ρ.

Problem 2

(a)

We have that the conditionally independent observation have the likelihood

$$f(\mathbf{y}|\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \theta_1^2 - \theta_2^2)^2\right)$$

with the prior distribution on $\boldsymbol{\theta}$ given as

$$f(\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2}\boldsymbol{\theta}^\top \Sigma^{-1}\boldsymbol{\theta}\right)$$

where the covariance matrix Σ is a known hyperparameter.

Bayes theorem then gives that the logarithm of the posterior can be written as

$$\ln f(\boldsymbol{\theta}|\mathbf{y}) = \ln \frac{f(\mathbf{y}|\boldsymbol{\theta})f(\boldsymbol{\theta})}{f(\mathbf{y})} = \ln f(\mathbf{y}|\boldsymbol{\theta}) + \ln f(\boldsymbol{\theta}) - \ln f(\mathbf{y}),$$

where $f(\mathbf{y})$ is the normalizing constant.

We thus have

$$\ln f(\boldsymbol{\theta}|\mathbf{y}) \propto -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \theta_1^2 - \theta_2^2)^2 - \frac{1}{2}\boldsymbol{\theta}^\top \Sigma^{-1}\boldsymbol{\theta} + C.$$

where C is the additive constant not depending on $\boldsymbol{\theta}$.

The gradient of the log posterior is then given as

$$\nabla_{\boldsymbol{\theta}} \ln f(\boldsymbol{\theta}|\mathbf{y}) = -\frac{1}{2\sigma^2}\left[2\sum_{i=1}^{n}(y_i - \theta_1^2 - \theta_2^2)\right](-2)\boldsymbol{\theta} - \Sigma^{-1}\boldsymbol{\theta} = \frac{2\boldsymbol{\theta}}{\sigma^2}\left[\sum_{i=1}^{n}(y_i - \theta_1^2 - \theta_2^2)\right] - \Sigma^{-1}\boldsymbol{\theta}$$

(b)

For the given standard deviance σ, the covariance matrix Σ, observations $\mathbf{y}$, Leapfrog step size ε and amount of steps L and amount of particles N, the HMC algorithm to sample $\boldsymbol{\theta}$ from $f(\boldsymbol{\theta}|\mathbf{y})$ is given by the following pseudocode:

Input: $\mathbf{y}, \sigma, \Sigma, \varepsilon, L, N$

Output: $\boldsymbol{\theta}_{1:N}$

1. Define $U(\boldsymbol{\theta}) = -\ln f(\boldsymbol{\theta}|\mathbf{y})$ and $K(\mathbf{v}) = \frac{1}{2}\texttt{sum}(\mathbf{v}.^2)$ such that $H(\boldsymbol{\theta}, \mathbf{v}) = U(\boldsymbol{\theta}) + K(\mathbf{v})$
2. Initialize the position $\boldsymbol{\theta}_1$ (Matlab indexing), preferably on circle.
3. `For` $n = 2, \ldots, N$
   (a) Sample $\mathbf{v_{n-1}} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{I}_2)$
   (b) Run Leapfrog for L steps with step size ε to get proposed $\mathbf{v}_n^*$ and $\boldsymbol{\theta}_n^*$:

$$[\mathbf{v}_n^*, \boldsymbol{\theta}_n^*] = \texttt{Leapfrog}(\boldsymbol{\theta}_{n-1}, \mathbf{v}_{n-1}, L, \varepsilon, \sigma^2, \Sigma, \mathbf{y})$$

   (c) Compute the proposed $H_n^* = H(\boldsymbol{\theta}_n^*, \mathbf{v}_n^*)$ and $H_{n-1} = H(\boldsymbol{\theta}_{n-1}, \mathbf{v}_{n-1})$ and accept $\boldsymbol{\theta}_n^*$ with probability

$$\alpha = \min\left[1, \exp\left(H_{n-1} - H_n^*\right)\right]$$

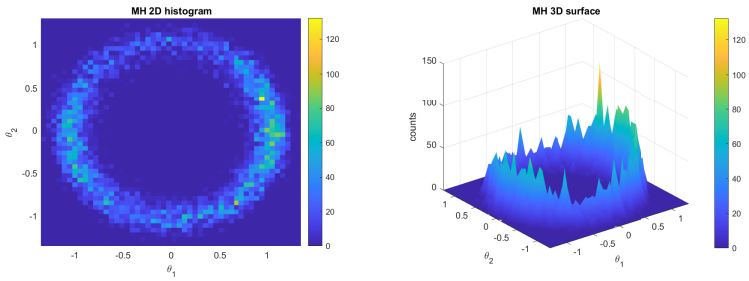
`end`

(c)

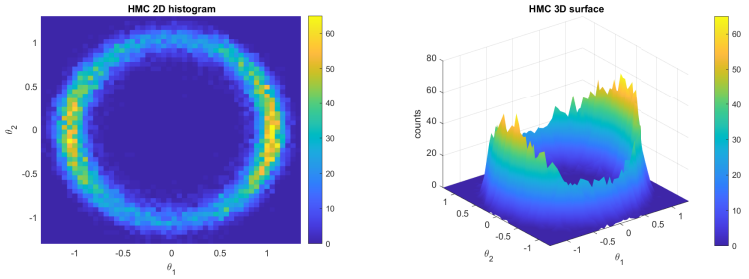
(i)

The recorded acceptance rate for MH was ≈ 0.29 in comparison to ≈ 0.98 for HMC. In Figure 4 we can see that the sample density from HMC resembles the true posterior density very well, while the MH does not do so as efficiently. This was the result for $\zeta = 0.4$ for MH and $L = 25$, $\epsilon = 0.05$ for HMC (More on this choice below in (ii)).

HMC achieves a higher acceptance rate and faster decay of the autocorrelation function than MH (as can be seen in Figure 5), at the cost of greater computational complexity. This is consistent with the theory where the MH algorithm tends to propose steps that go outside the circle, if one does not decrease the step size a lot, leading to more rejections. On the other hand, HMC uses the gradient of the potential to guide the proposed steps along the high density regions, leading to higher acceptance rates.



(a) 2D histogram for MH (b) 3D histogram for MH



(c) 2D histogram for HMC (d) 3D histogram for HMC

Figure 4: Sample densities for MH and HMC

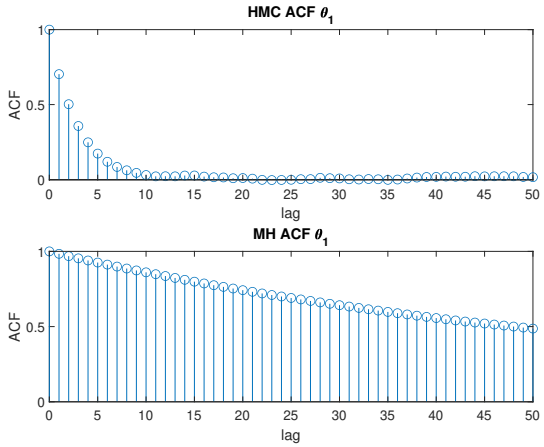


Figure 5: Autocorrelation function for HMC and MH. (θ_1)

(ii)

When investigating the performance of MH under varying ζ, small values produced high acceptance rates but slow movement through the posterior, while large values caused most proposals to be rejected. A value of $\zeta = 0.4$ was found to give a good balance, yielding an acceptance rate within the theoretically recommended range of 23%–44%.

For HMC, small ε and large L keeps the Leapfrog algorithm accurate but slows exploration around the circle, while large ε and small L causes most proposals to be rejected. One needs to balance ε with L to achieve a desirable acceptance rate. We found $\varepsilon = 0.05$ and $L = 25$ to work well in practice.